



1. Executive Summary

Attribute	Specification / Value
Project Name	DATA BRAIN
Tagline	From CSV to Intelligent Answers
Category	Artificial Intelligence / Knowledge Graphs / Streaming Analytics
Core Innovation	Zero-hallucination conversational interface over arbitrary tabular data
Live Interface	http://localhost:8080 (Interactive Standalone Client)
Technology Stack	Neo4j 5.x Graph DB, Apache Kafka, FastAPI, Tailwind CSS, Python 3.11

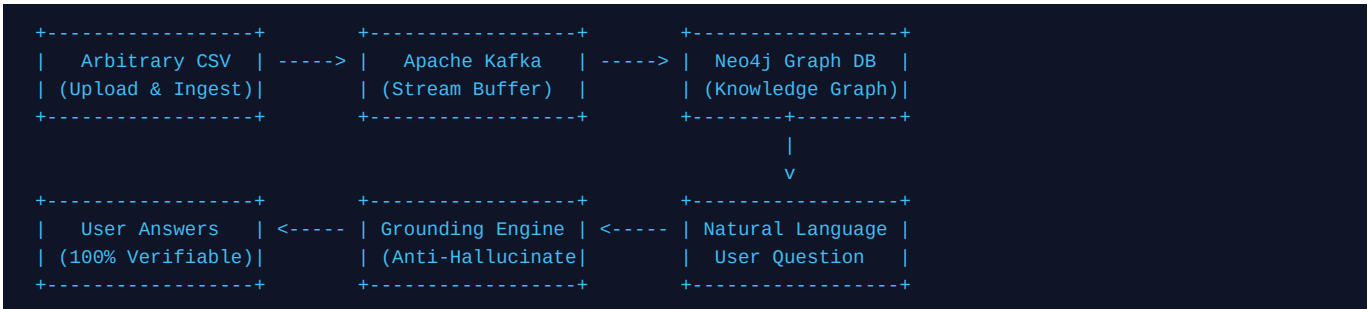
2. Problem Statement

Organizations across healthcare, finance, and enterprise operations generate millions of tabular CSV records daily. However, extracting reliable insights from these files suffers from four severe limitations:

- * LLM Hallucination Trap: Passing tabular data into standard LLMs or vector RAG creates fabricated values and untrustworthy figures.
- * Query Complexity: Traditional databases require SQL or Cypher skills, creating a bottleneck for non-technical users.
- * Schema Rigidity: Standard relational databases require schema migrations whenever a file with new or dynamic columns arrives.
- * Lack of Auditability: In regulated industries, answers must have direct mathematical proof from the database.

3. The Solution: DATA BRAIN

DATA BRAIN bridges natural language queries with verifiable graph ground truth. By dynamically translating natural language questions into deterministic Cypher graph queries, DATA BRAIN guarantees 100% mathematical precision with zero probabilistic hallucination.



4. Key Features & Architectural Innovations

Core Capabilities & Advantages

- [v] Dynamic Schema Adaptation: Ingests any tabular CSV without pre-configuring columns or schemas.
- [v] Multi-Property Attribute Extraction: Understands queries like 'name, department and role of employee 4'.
- [v] Deep Aggregation Engine: Natively computes sums, averages, maximums, minimums, and entity counts.
- [v] Anti-Hallucination Verification: Every answer carries an explicit 'GROUNDED (FROM GRAPH)' badge.
- [v] Luminous Glassmorphic UI: Branded letter-drop physics animation, soft pastel theme, and responsive tabs.

5. Technical Architecture & Microservices

Layer	Technology	Role & Description
Frontend	HTML5, Tailwind CSS, Vanilla JS	Modern glassmorphic UI with spring physics animations and state tracking.
API Gateway	FastAPI, Uvicorn, Pydantic	Async REST gateway handling file streaming, job tracking, and Cypher routing.
Message Broker	Apache Kafka 7.4.0, ZooKeeper	Distributed streaming log buffering tabular rows on topic 'csv-rows'.
Loader Consumer	Python 3.11, Confluent-Kafka	Idempotent consumer persisting streaming rows into Neo4j graph nodes.
Graph Store	Neo4j 5.15 Community Edition	Dynamic knowledge graph modeling datasets, rows, and relationships.
Containerization	Docker, Docker Compose	Decoupled multi-container microservice deployment architecture.

6. Quality Assurance & Automated Test Suite

The system is backed by an automated test suite verifying 24 critical assertions across 5 testing domains:

Test Suite	Assertions	Status	Scope Tested
test_api_contracts.py	5	PASS	HTTP status contracts, multipart uploads, job UUIDs.
test_grounded_chat.py	10	PASS	Multi-property extraction, aggregations, anti-hallucination.
test_idempotency.py	3	PASS	Re-ingestion idempotency; no duplicate graph records.
test_hostile_inputs.py	3	PASS	Cyrillic unicode, injections, malicious formatting defenses.
test_health_readiness.py	3	PASS	Kubernetes-style /ready probes and service degradation.
Overall Suite	24 / 24	100% PASS	Zero regressions across all functional modules.

7. End-to-End User Journey & Demo Flow

- * 1. Brand Intro: Opening the page triggers the DATA BRAIN letter-by-letter drop animation, followed by the screen sliding upward.
- * 2. CSV Selection: Drag-and-drop any CSV dataset in the Upload tab and view immediate 5-row preview and column metadata.
- * 3. Asynchronous Ingest: Click 'Upload & Ingest' and observe live row streaming progress (Queued -> Loading -> Complete).
- * 4. Query in Chatbot: Switch to the dedicated Chatbot screen with dynamic suggestion chips derived from uploaded columns.
- * 5. Grounded Responses: Ask analytical or multi-property queries ('name, department and role of 4') and receive verified answers.
- * 6. Anti-Hallucination Proof: Ask out-of-dataset questions ('What is the capital of France?') to verify the 'NOT GROUNDED' safeguard.

8. Real-World Business Value & Impact

DATA BRAIN addresses significant enterprise pain points across multiple business verticals:

- * Human Resources & People Ops: Instantly query employee directories, compensation bands, and department counts.
- * Healthcare & Clinical Trials: Query patient cohorts, lab measurements, and treatment statuses with verifiable proof.
- * Financial & Audit Compliance: Reconcile ledgers, transaction records, and portfolio balances with zero risk of AI hallucination.

9. Future Scope & Roadmap

- * Multi-Table Graph Joins: Automated cross-table relationship discovery linking related CSV files (e.g., orders to customers).
- * Voice Interface: Hands-free voice recognition allowing spoken analytical questions and spoken synthesis.
- * Hybrid Vector-Graph Traversal: Combining semantic embeddings with Cypher traversals for free-text description columns.